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Lackner et al.

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(54) **SYSTEM AND METHOD FOR PASSIVE
COLLECTION OF ATMOSPHERIC CARBON
DIOXIDE WITH ENHANCED AIR FLOW**

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2259/40088; B01D 2259/40094;
(Continued)

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(56)

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(57)

ABSTRACT

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3, 2022.

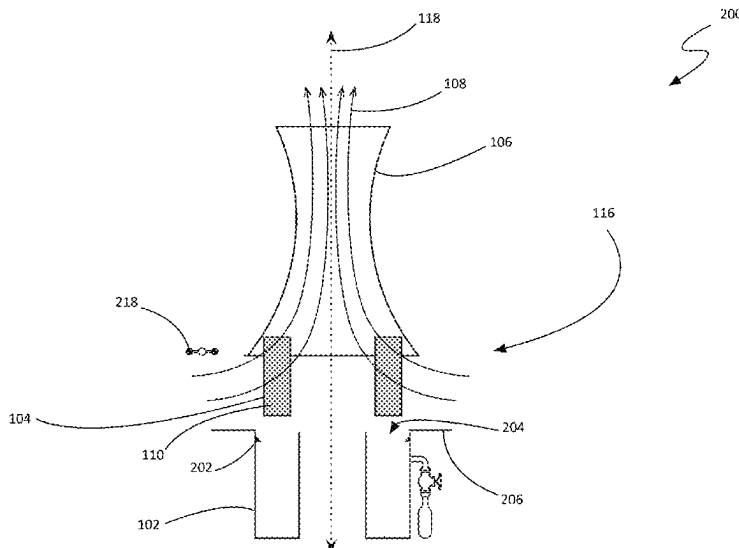
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B01D 53/04 (2006.01)
F24S 70/60 (2018.01)

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CPC **B01D 53/0438** (2013.01); **B01D 53/0446**
(2013.01); **F24S 70/60** (2018.05); **B01D**
2257/504 (2013.01); **B01D 2258/06** (2013.01);
B01D 2259/40098 (2013.01)

(58) **Field of Classification Search**
CPC B01D 53/04; B01D 53/0438; B01D

A carbon dioxide collection system having a release enclosure, a capture structure, and a chimney is disclosed. The release enclosure includes a sorbent regeneration system. The capture structure includes a sorbent material, and is movable between collection and release configurations. The chimney is shaped such that an airflow upward through the chimney is created. The chimney is positioned above the release enclosure such that the airflow passes through the capture structure while in the collection configuration. The collection configuration includes the capture structure being elevated above the release enclosure so the sorbent material is exposed to the airflow generated by the chimney, allowing the sorbent material to capture CO₂ from the airflow. The release configuration includes the capture structure being sufficiently enclosed inside the release enclosure that the sorbent regeneration system may operate on the sorbent material to release CO₂ collected by the capture structure to form an enriched fluid.

20 Claims, 14 Drawing Sheets



(58) **Field of Classification Search**

CPC B01D 2259/40096; Y02C 20/40; F24S
70/60; F24S 70/30
USPC 423/230; 96/121, 123, 126, 143, 150;
95/107, 139

See application file for complete search history.

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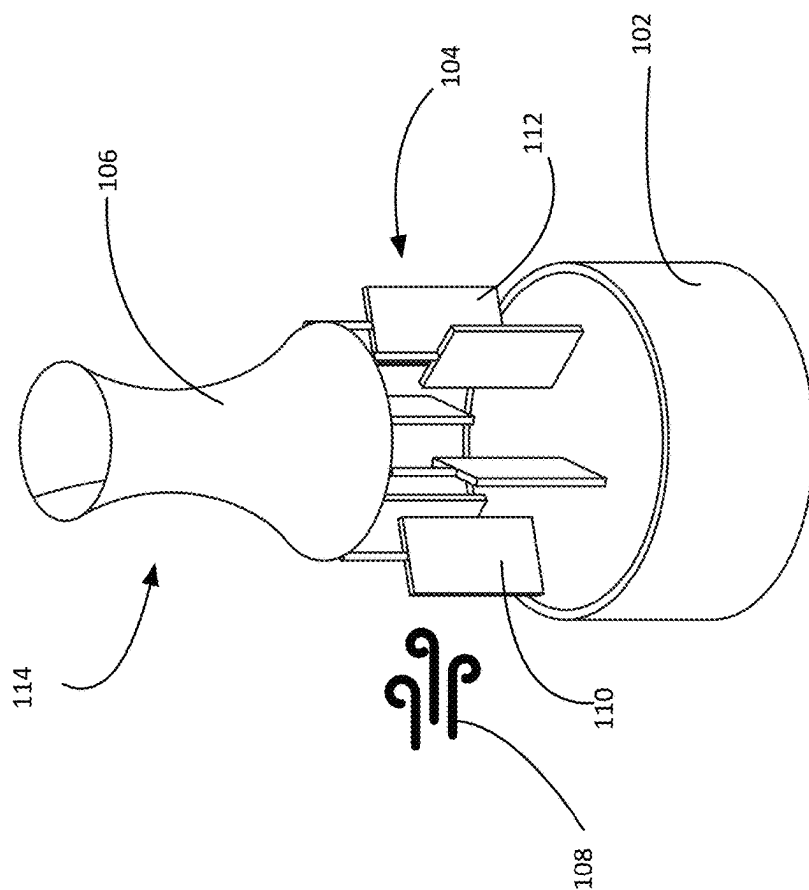


FIG. 1A

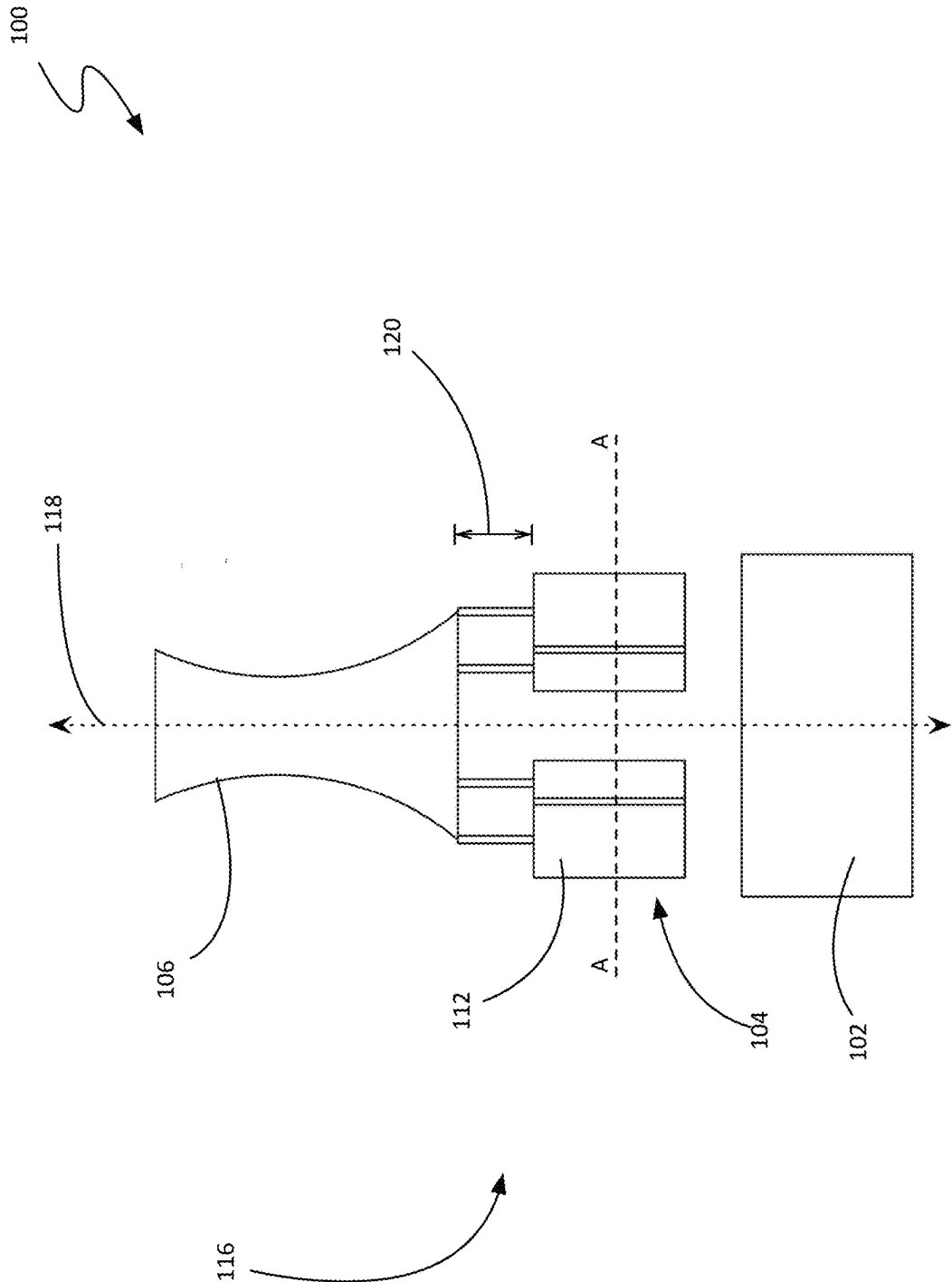


FIG. 1B

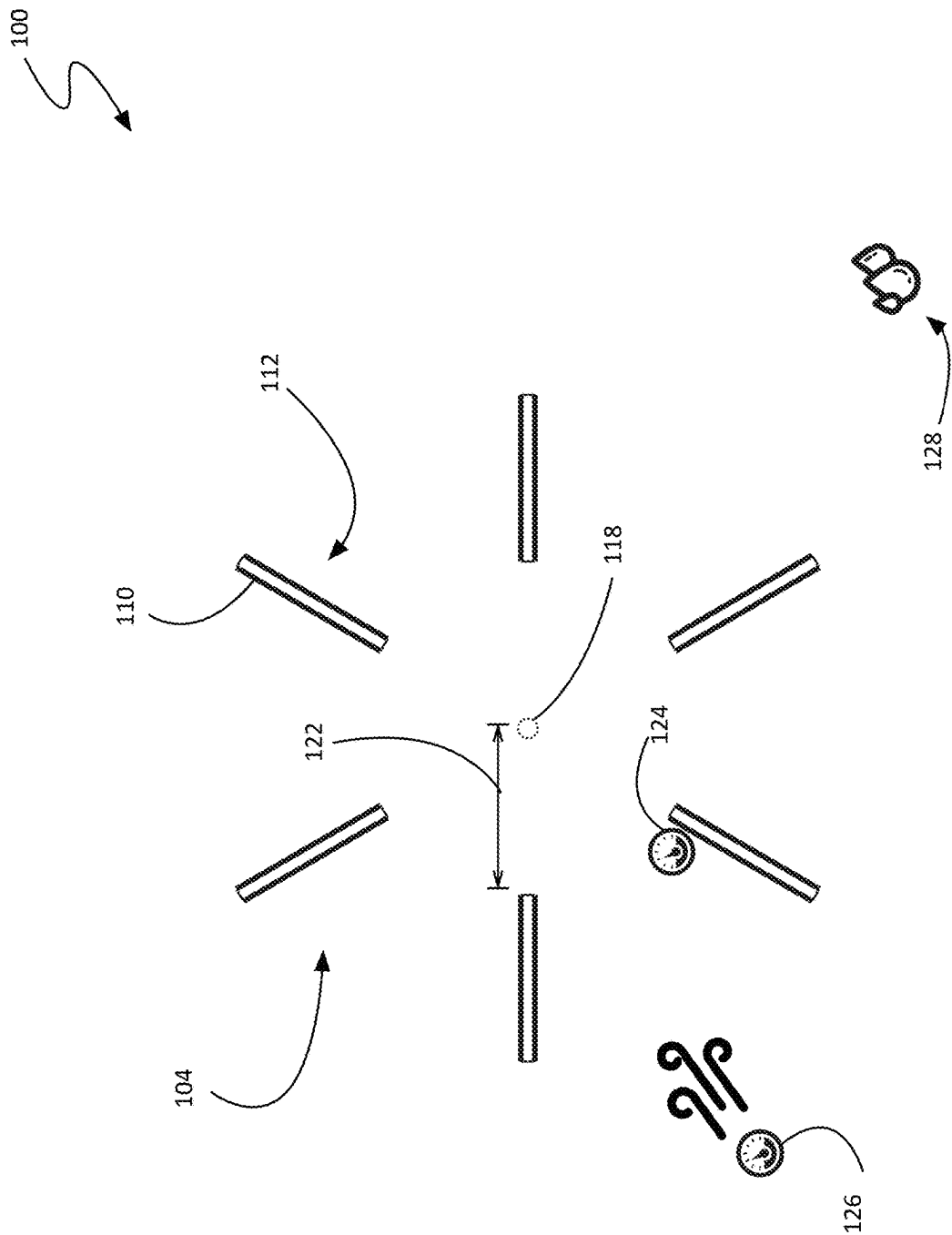


FIG. 1C

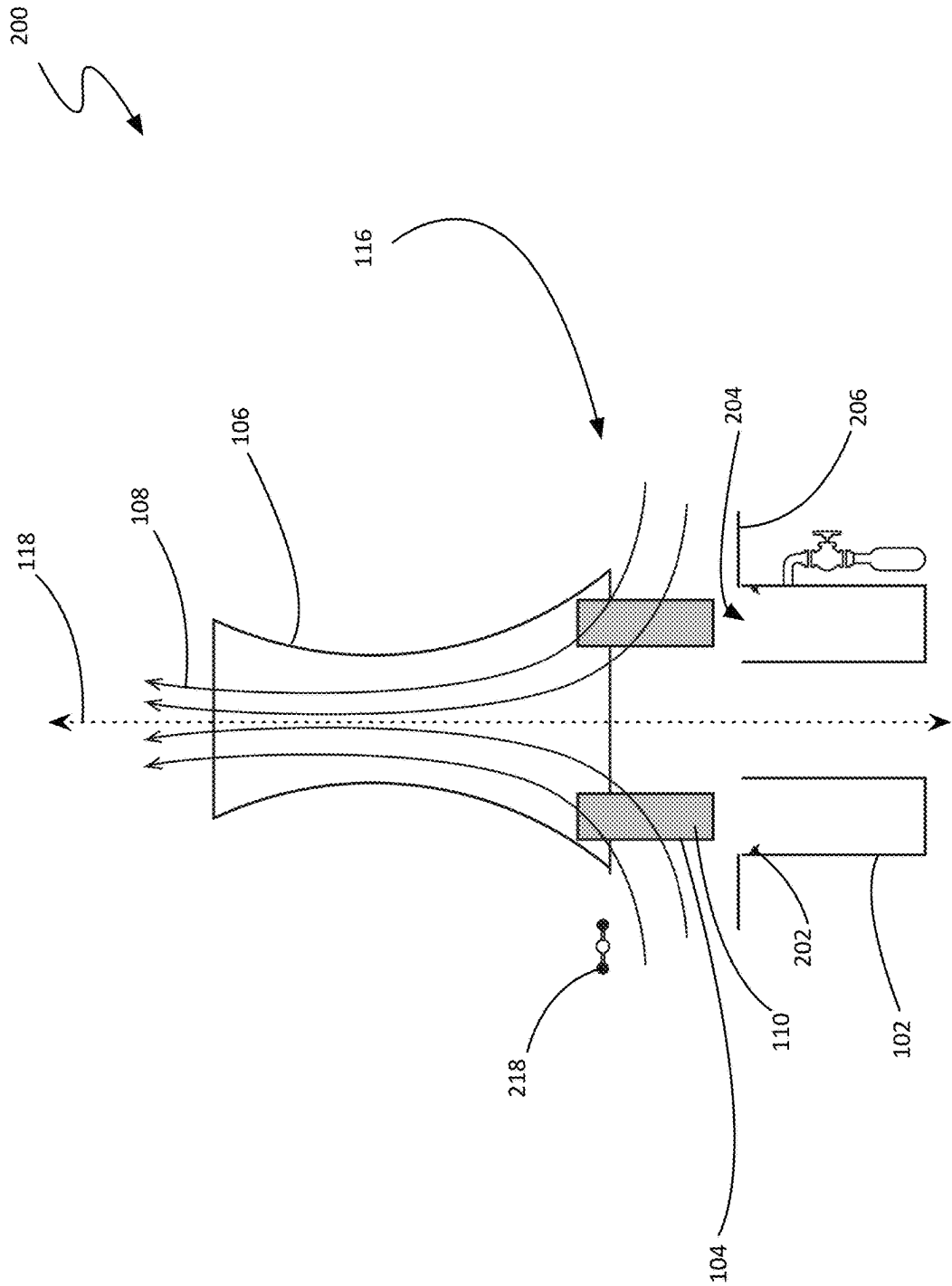


FIG. 2A

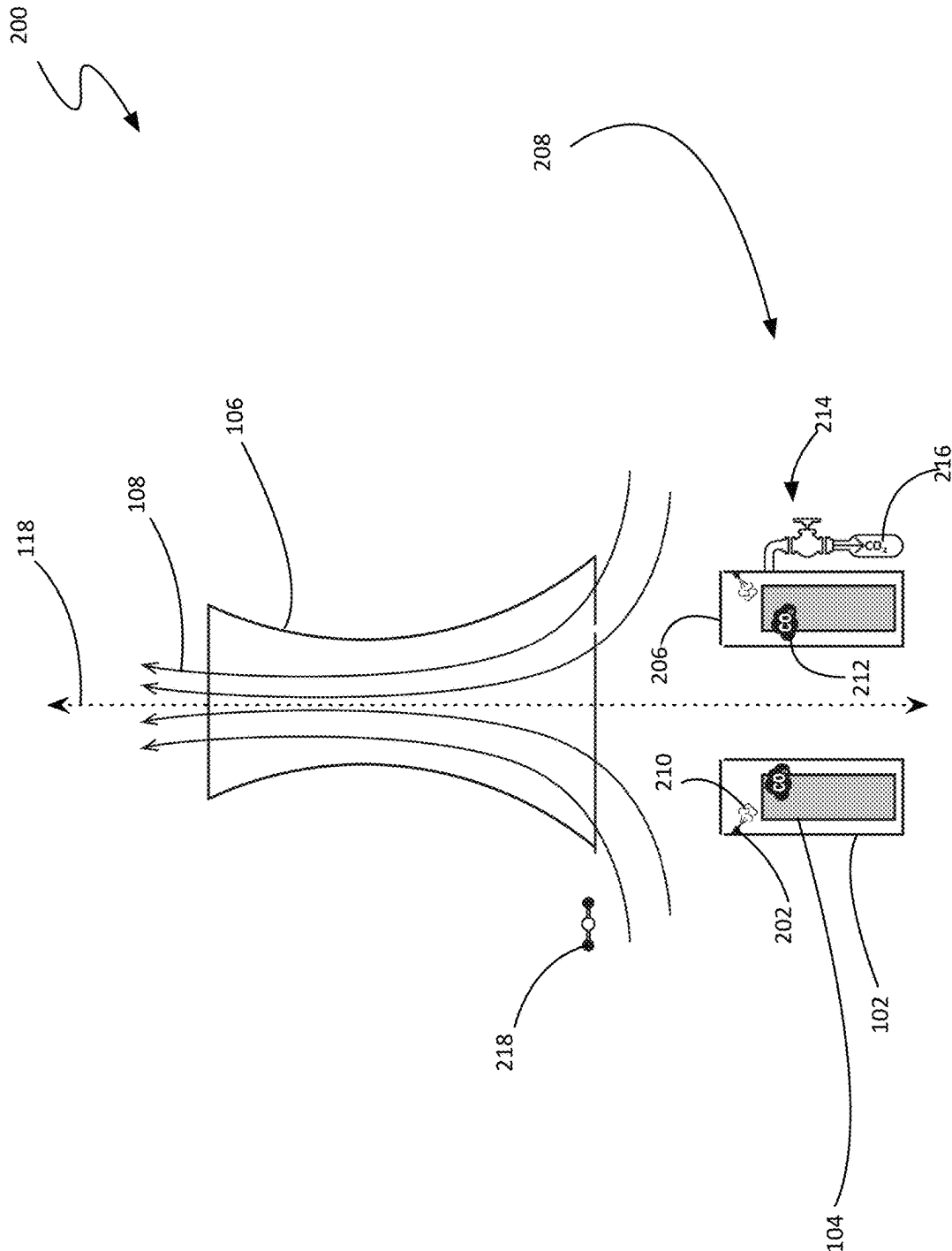


FIG. 2B

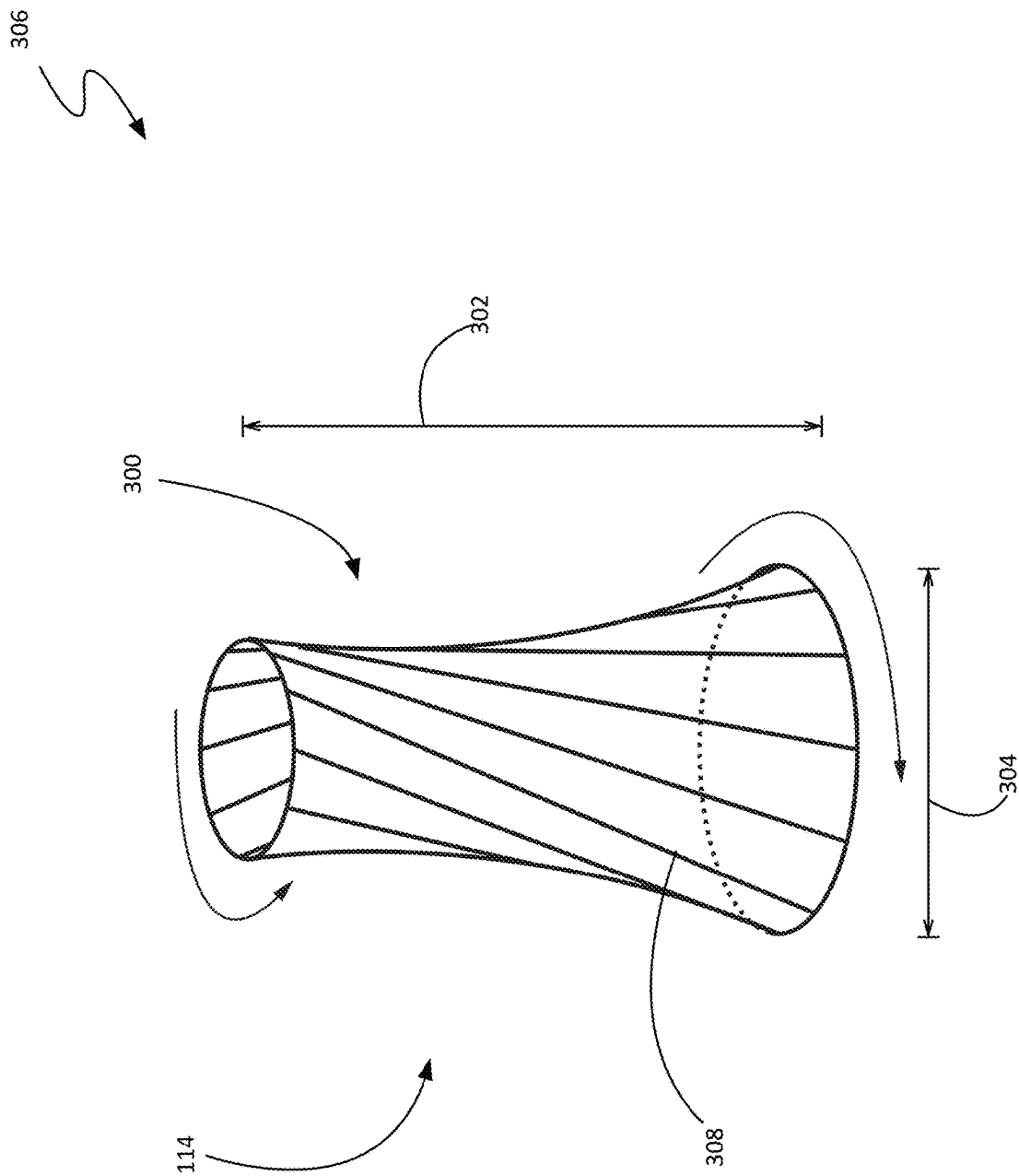


FIG. 3

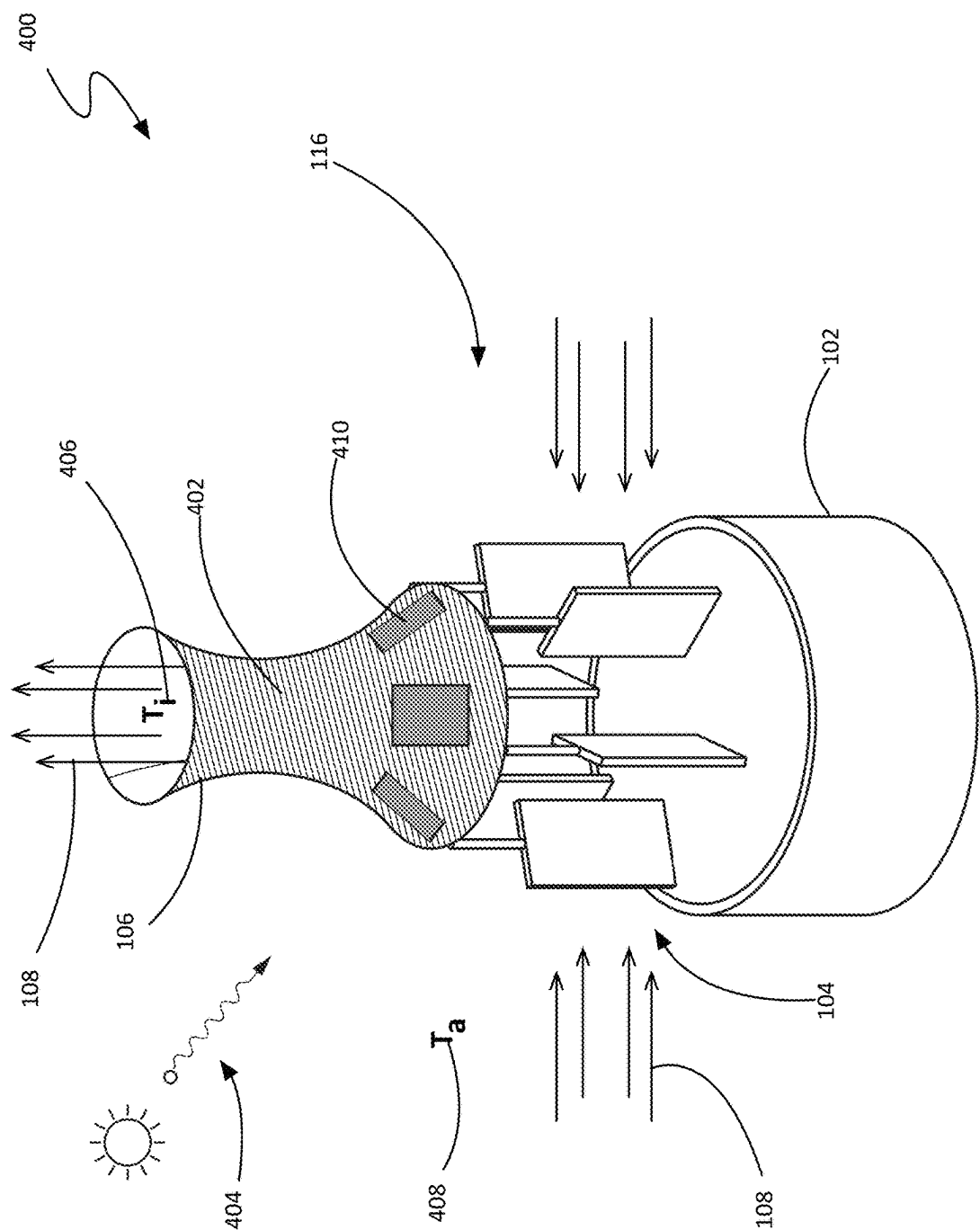


FIG. 4

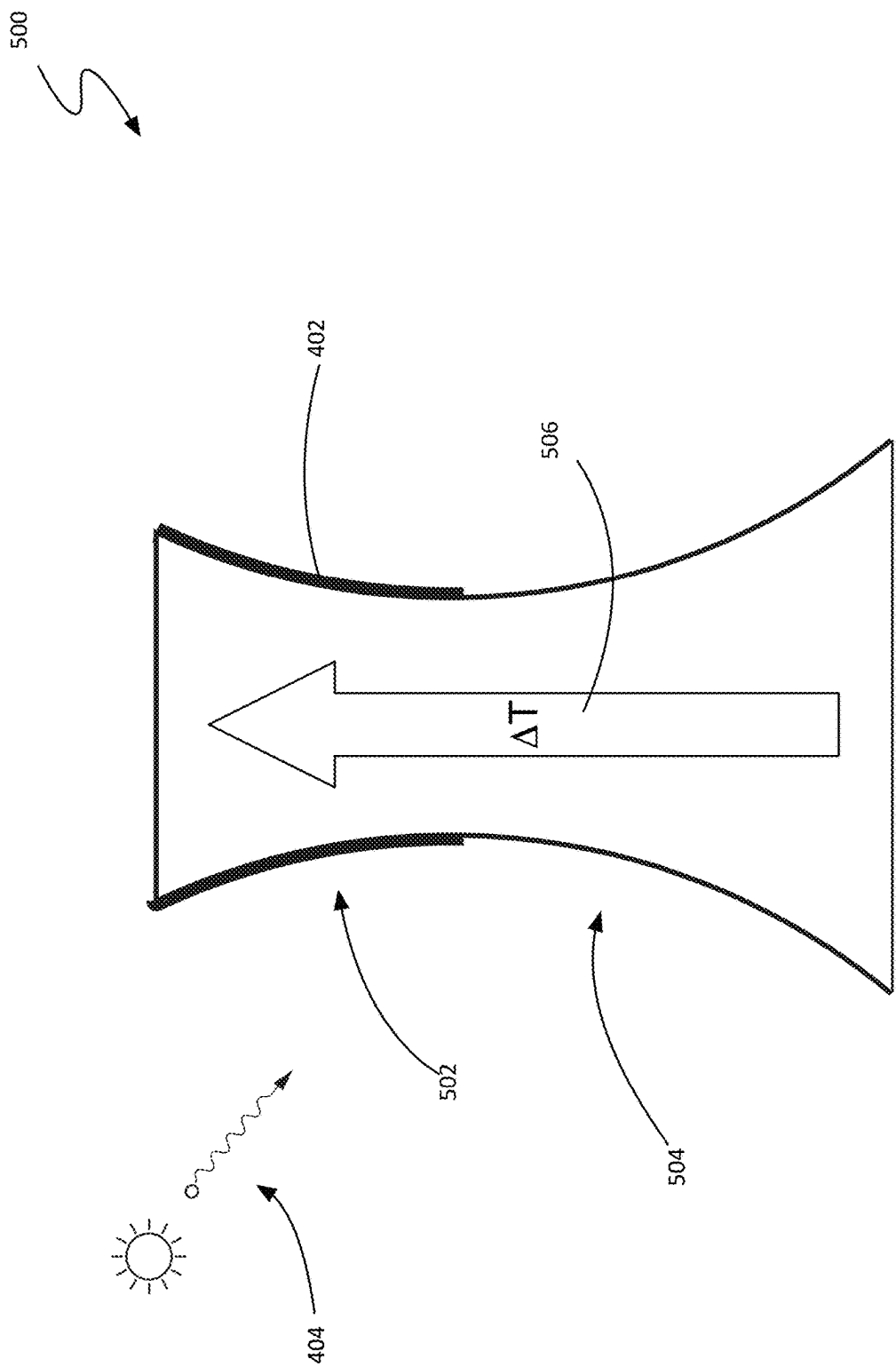


FIG. 5

600

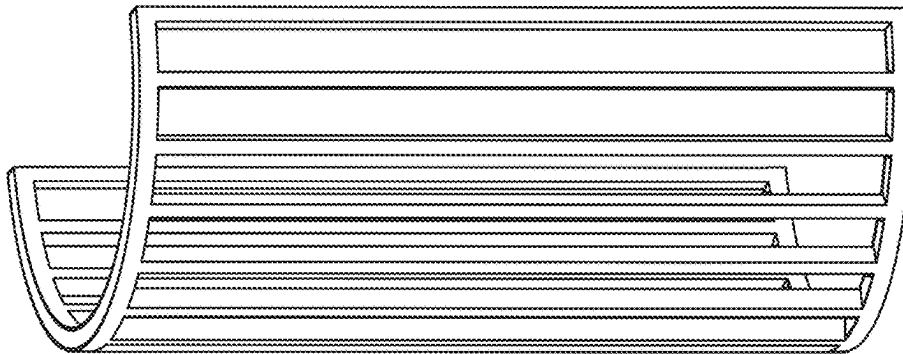
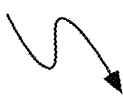


FIG. 6

700

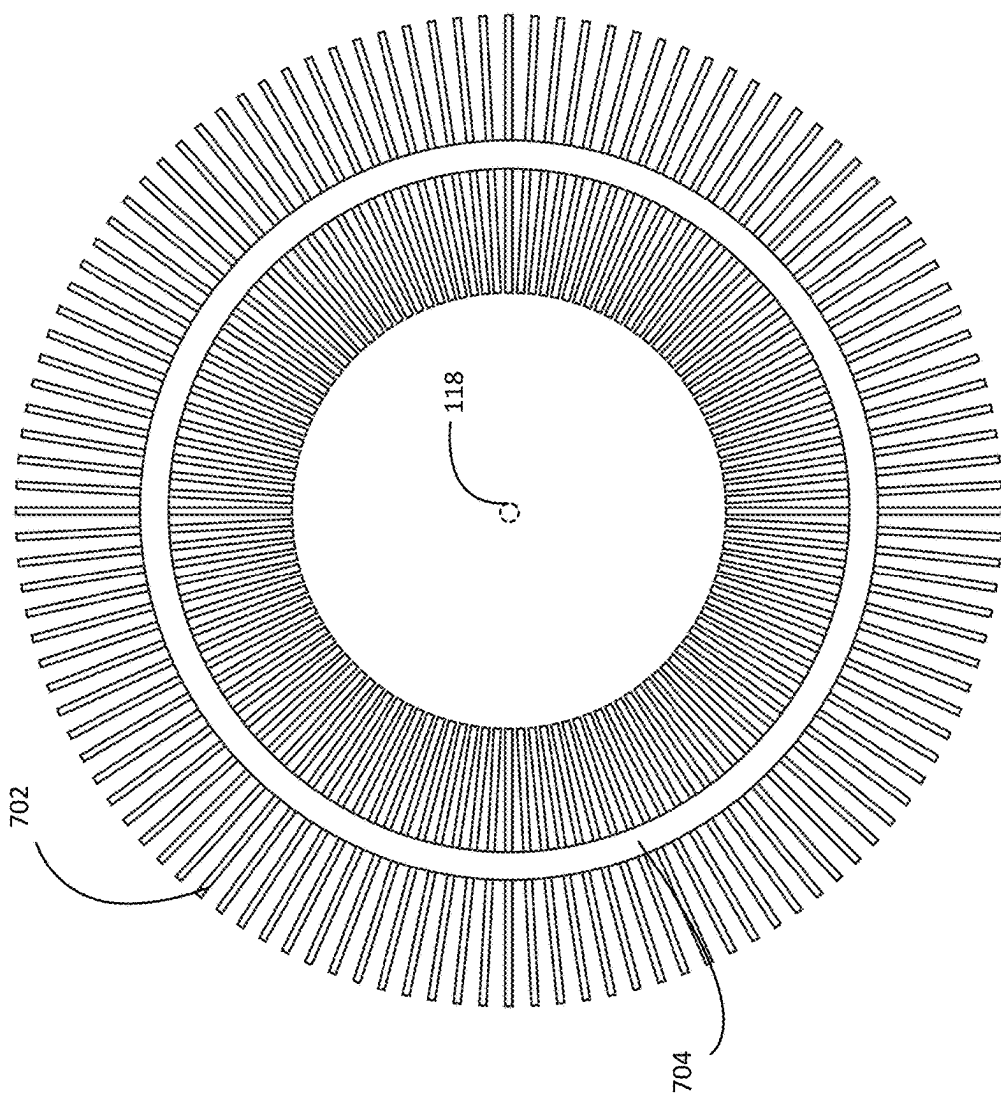


FIG. 7

800

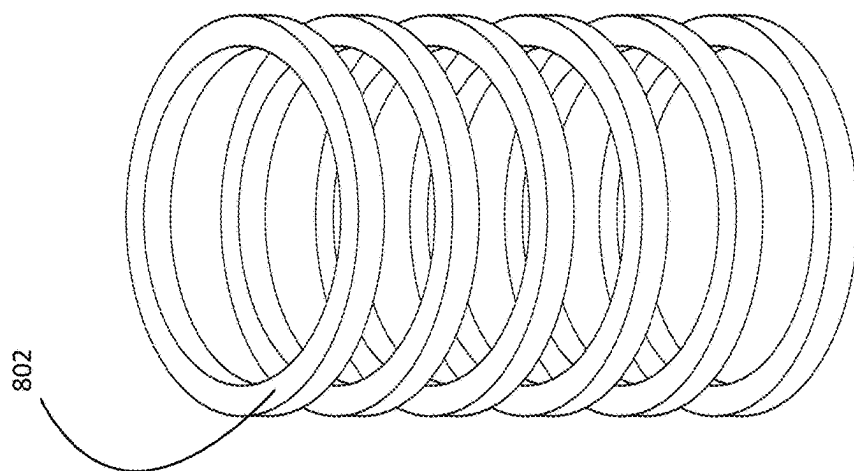
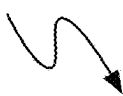


FIG. 8

900

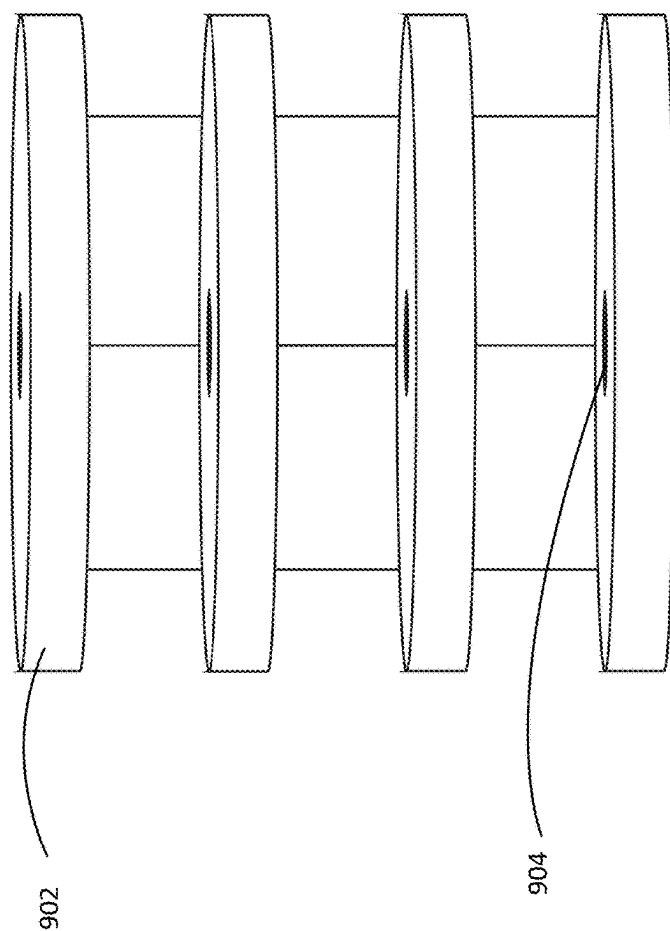


FIG. 9

1000

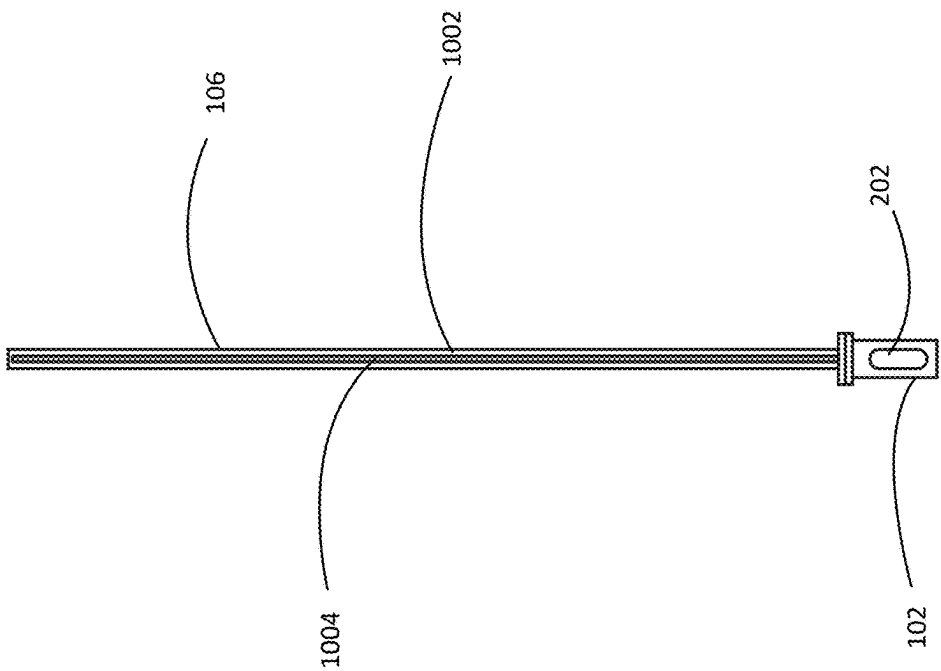


FIG. 10

1100

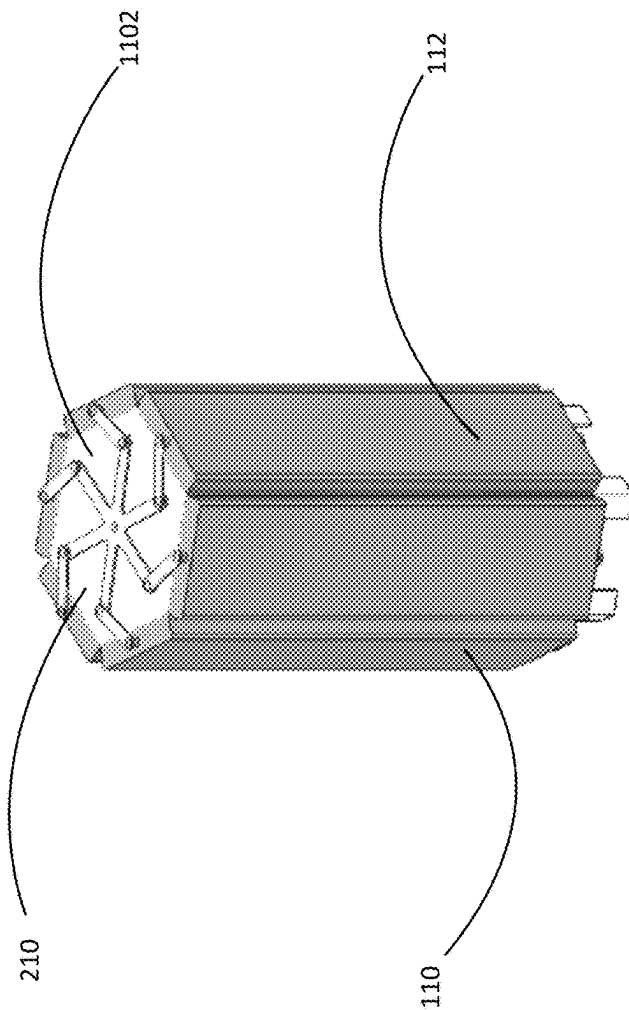


FIG. 11

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SYSTEM AND METHOD FOR PASSIVE COLLECTION OF ATMOSPHERIC CARBON DIOXIDE WITH ENHANCED AIR FLOW

RELATED APPLICATIONS

This application claims the benefit of U.S. provisional patent application No. 63/316,082, filed Mar. 3, 2022, titled "System and Method for Passive Collection of Atmospheric Carbon Dioxide with Enhanced Air Flow," the entirety of the disclosure of which is hereby incorporated by this reference.

TECHNICAL FIELD

Aspects of this document relate generally to the collection of atmospheric carbon dioxide.

BACKGROUND

The need for technologies to remove carbon dioxide from ambient air has been well established. In addition to conservation, reduced-carbon processes, and on-site capture efforts, a significant amount of carbon dioxide will need to be removed from the atmosphere to avoid a looming climate change crisis.

Capture of carbon dioxide from ambient air at an affordable price could become a critical tool in managing the anthropogenic carbon cycle. Air capture technology would greatly enhance the options for developing the world's energy infrastructure and would be a major asset in the fight against climate change. Combined with a carbon dioxide (CO₂) storage technology, air capture of CO₂ could compensate for CO₂ emissions from any source, without requiring changes to the existing infrastructure and without requiring proximity to the point of emission. Air capture technology makes it possible for existing infrastructures to live out their natural life spans, and it permits the continued use of carbon-based fuels in distributed and mobile applications, for example in the transportation sector. With air capture technology, it is possible to continue the use of liquid hydrocarbon fuels while eliminating their climate impact.

Nevertheless, these technologies are still new and the early air capture processes require large amounts of energy to operate. Since the carbon dioxide in the ambient air is very dilute, atmospheric CO₂ collectors can quickly overrun a tight energy budget for drawing in and processing air in bulk.

A promising technology that is well adapted for capturing dilute atmospheric carbon dioxide in an energy efficient manner is direct air capture, or DAC. There exist two types of DAC regarding airflow; active, which uses large fans to bring the CO₂ to the sorbents, and passive, which takes advantage of natural air currents like wind. The current capture cost of DAC (500-600 USD/ton CO₂) is too expensive, mainly due to operational expenses. Although the distinct costs associated with active capture (e.g., the operation of fans, etc.) are presumably low, the costs are relatively fixed since fan technology has probably reached its technical limit. On the other hand, passive DAC devices can have lower operating expenses but tend to have a larger cross-sectional area than active devices, since they need to expose a larger surface area of sorbent to compensate for the intermittent nature of ambient air flows.

Direct air capture generally constitutes both CO₂ capture and its release. The release process is considered the primary economic limiting factor, driven by either temperature,

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pressure, moisture, electrochemical gradient, or combination of these drivers. In particular, the moisture-swing process requires relatively minor amounts of energy, in which the energy to release the CO₂ is the latent heat of the water. However, this process consumes significant amounts of water relative to the capture tonnage.

The limitations of active capture (i.e., fan driven capture) is the cost of the fans and motors and their operational cost. The limitation of passive capture is the cost of the required increase in surface area of exposed sorbent. The benefits of each method come at a cost.

SUMMARY

According to one aspect, a carbon dioxide collection system includes a release enclosure having an opening and a sorbent regeneration system. The system also includes a capture structure having a plurality of vertical panels oriented radially about a central axis of the capture structure, with each vertical panel having a sorbent material. The capture structure is movable between a collection configuration and a release configuration. The system includes a chimney having a hyperboloid structure and shaped such that an airflow upward through the chimney is created, with the chimney positioned above the release enclosure such that the airflow passes through the capture structure before exiting the chimney while the capture structure is in the collection configuration. The chimney has a heating material configured to absorb solar radiation such that an air temperature inside the chimney is greater than an ambient air temperature outside the chimney. The system also includes at least one solar panel coupled to the chimney, and a product outlet in fluid communication with the inside of the release enclosure and configured to emit a product stream of enriched fluid. The collection configuration includes the capture structure being elevated above the release enclosure such that the sorbent material of the capture structure is exposed to the airflow generated by the chimney, allowing the sorbent material to capture carbon dioxide from the airflow. The release configuration includes the capture structure being sufficiently enclosed inside the release enclosure that the sorbent regeneration system may operate on the sorbent material of the capture structure to release carbon dioxide collected by the capture structure while in the collection configuration to form an enriched fluid within the release enclosure.

Particular embodiments may comprise one or more of the following features. The collection configuration may include at least one of the capture structure being separated from the chimney by at least a vertical separation and/or the sorbent material of the capture structure being distanced from a central axis of the capture structure by at least a radius. At least one of the vertical separation and the radius may be chosen such that a CO₂ capture rate of the sorbent material as arranged in the capture structure is substantially equal to a CO₂ exposure rate caused by the airflow passing through the capture structure while the capture structure is in the collection configuration. The at least one of the radius and the vertical separation may be able to be modified after the carbon dioxide collection system has been installed, in response to a change in an environmental condition. The release enclosure may further include a lid sized to cover the opening and configured to seal the capture structure inside the release enclosure when in the release configuration. The chimney may be one of a cooling tower and a smokestack. The chimney may include a dynamic geometry such that at least one of a height of the chimney and a width of the

chimney may be modified after manufacture. The heating material may be confined to an upper portion of the chimney such that the upper portion absorbs more solar radiation than a lower portion of the chimney, thereby creating a thermal gradient to enhance the airflow through the chimney. The collection configuration may include the capture structure being at least partially inside of the chimney.

According to another aspect of the disclosure, a carbon dioxide collection system includes a release enclosure having an opening and a sorbent regeneration system. The system also includes a capture structure having a sorbent material. The capture structure is movable between a collection configuration and a release configuration. The system includes a chimney shaped such that an airflow upward through the chimney is created, with the chimney positioned above the release enclosure such that the airflow passes through the capture structure before exiting the chimney while the capture structure is in the collection configuration. The system also includes a product outlet in fluid communication with the inside of the release enclosure and configured to emit a product stream of enriched fluid. The collection configuration includes the capture structure being elevated above the release enclosure such that the sorbent material of the capture structure is exposed to the airflow generated by the chimney, allowing the sorbent material to capture carbon dioxide from the airflow. The release configuration includes the capture structure being sufficiently enclosed inside the release enclosure that the sorbent regeneration system may operate on the sorbent material of the capture structure to release carbon dioxide collected by the capture structure while in the collection configuration to form an enriched fluid within the release enclosure.

Particular embodiments may comprise one or more of the following features. The capture structure may include a plurality of vertical panels oriented radially about a central axis of the capture structure. Each vertical panel may include the sorbent material. The collection configuration may include at least one of the capture structure being separated from the chimney by at least a vertical separation and the sorbent material of the capture structure being distanced from a central axis of the capture structure by at least a radius. At least one of the vertical separation and the radius may be chosen such that a CO₂ capture rate of the sorbent material as arranged in the capture structure is substantially equal to a CO₂ exposure rate caused by the airflow passing through the capture structure while the capture structure is in the collection configuration. The at least one of the radius and the vertical separation may be able to be modified after the carbon dioxide collection system has been installed, in response to a change in an environmental condition. The release enclosure may further include a lid sized to cover the opening and configured to seal the capture structure inside the release enclosure when in the release configuration. At least one solar panel may be coupled to the chimney. The chimney may be one of a cooling tower and a smokestack. The chimney may be a hyperboloid structure. The chimney may include a dynamic geometry such that at least one of a height of the chimney and a width of the chimney may be modified after manufacture. The chimney may include a heating material configured to absorb solar radiation such that an air temperature inside the chimney may be greater than an ambient air temperature outside the chimney. The heating material may be confined to an upper portion of the chimney such that the upper portion absorbs more solar radiation than a lower portion of the chimney, thereby creating a thermal gradient to enhance the airflow through

the chimney. The collection configuration may include the capture structure being at least partially inside of the chimney.

Aspects and applications of the disclosure presented here are described below in the drawings and detailed description. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts. The inventors are fully aware that they can be their own lexicographers if desired. The inventors expressly elect, as their own lexicographers, to use only the plain and ordinary meaning of terms in the specification and claims unless they clearly state otherwise and then further, expressly set forth the “special” definition of that term and explain how it differs from the plain and ordinary meaning. Absent such clear statements of intent to apply a “special” definition, it is the inventors’ intent and desire that the simple, plain and ordinary meaning to the terms be applied to the interpretation of the specification and claims.

The inventors are also aware of the normal precepts of English grammar. Thus, if a noun, term, or phrase is intended to be further characterized, specified, or narrowed in some way, then such noun, term, or phrase will expressly include additional adjectives, descriptive terms, or other modifiers in accordance with the normal precepts of English grammar. Absent the use of such adjectives, descriptive terms, or modifiers, it is the intent that such nouns, terms, or phrases be given their plain, and ordinary English meaning to those skilled in the applicable arts as set forth above.

Further, the inventors are fully informed of the standards and application of the special provisions of 35 U.S.C. § 112(f). Thus, the use of the words “function,” “means” or “step” in the Detailed Description or Description of the Drawings or claims is not intended to somehow indicate a desire to invoke the special provisions of 35 U.S.C. § 112(f), to define the invention. To the contrary, if the provisions of 35 U.S.C. § 112(f) are sought to be invoked to define the inventions, the claims will specifically and expressly state the exact phrases “means for” or “step for”, and will also recite the word “function” (i.e., will state “means for performing the function of [insert function]”), without also reciting in such phrases any structure, material or act in support of the function. Thus, even when the claims recite a “means for performing the function of . . .” or “step for performing the function of . . .,” if the claims also recite any structure, material or acts in support of that means or step, or that perform the recited function, then it is the clear intention of the inventors not to invoke the provisions of 35 U.S.C. § 112(f). Moreover, even if the provisions of 35 U.S.C. § 112(f) are invoked to define the claimed aspects, it is intended that these aspects not be limited only to the specific structure, material or acts that are described in the preferred embodiments, but in addition, include any and all structures, materials or acts that perform the claimed function as described in alternative embodiments or forms of the disclosure, or that are well known present or later-developed, equivalent structures, material or acts for performing the claimed function.

The foregoing and other aspects, features, and advantages will be apparent to those artisans of ordinary skill in the art from the DESCRIPTION and DRAWINGS, and from the CLAIMS.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will hereinafter be described in conjunction with the appended drawings, where like designations denote like elements, and:

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FIGS. 1A-1C are perspective, side, and cross-sectional views of a carbon dioxide collection system with enhanced air flow, respectively;

FIGS. 2A and 2B are cross-sectional views of a carbon dioxide collection system with a capture structure in the collection and release configurations, respectively;

FIG. 3 is a perspective view of a chimney with dynamic geometry;

FIG. 4 is a perspective view of a carbon dioxide collection system using solar heating;

FIG. 5 is a cross-sectional view of a chimney using a heating material to create a thermal gradient;

FIG. 6 is a front view of a portion of a capture structure without sorbent material;

FIG. 7 is a top view of a capture structure with a capture rack;

FIG. 8 is a perspective view of a capture structure with a plurality of rings;

FIG. 9 is a side view of a capture structure with a plurality of disks;

FIG. 10 is a cross-sectional view of a carbon dioxide collection system with an internal capture structure; and

FIG. 11 is a perspective view of an intermittent release enclosure.

DETAILED DESCRIPTION

This disclosure, its aspects and implementations, are not limited to the specific material types, components, methods, or other examples disclosed herein. Many additional material types, components, methods, and procedures known in the art are contemplated for use with particular implementations from this disclosure. Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any components, models, types, materials, versions, quantities, and/or the like as is known in the art for such systems and implementing components, consistent with the intended operation.

The word “exemplary,” “example,” or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It is to be appreciated that a myriad of additional or alternate examples of varying scope could have been presented, but have been omitted for purposes of brevity.

While this disclosure includes a number of embodiments in many different forms, there is shown in the drawings and will herein be described in detail particular embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the disclosed methods and systems, and is not intended to limit the broad aspect of the disclosed concepts to the embodiments illustrated.

The need for technologies to remove carbon dioxide from ambient air has been well established. In addition to conservation, reduced-carbon processes, and on-site capture efforts, a significant amount of carbon dioxide will need to be removed from the atmosphere to avoid a looming climate change crisis.

Capture of carbon dioxide from ambient air at an affordable price could become a critical tool in managing the

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anthropogenic carbon cycle. Air capture technology would greatly enhance the options for developing the world's energy infrastructure and would be a major asset in the fight against climate change. Combined with a carbon dioxide (CO₂) storage technology, air capture of CO₂ could compensate for CO₂ emissions from any source, without requiring changes to the existing infrastructure and without requiring proximity to the point of emission. Air capture technology makes it possible for existing infrastructures to live out their natural life spans, and it permits the continued use of carbon-based fuels in distributed and mobile applications, for example in the transportation sector. With air capture technology, it is possible to continue the use of liquid hydrocarbon fuels while eliminating their climate impact.

Nevertheless, these technologies are still new and the early air capture processes require large amounts of energy to operate. Since the carbon dioxide in the ambient air is very dilute, atmospheric CO₂ collectors can quickly overrun a tight energy budget for drawing in and processing air in bulk.

A promising technology that is well adapted for capturing dilute atmospheric carbon dioxide in an energy efficient manner is direct air capture, or DAC. There exist two types of DAC regarding airflow; active, which uses large fans to bring the CO₂ to the sorbents, and passive, which takes advantage of natural air currents like wind. The current capture cost of DAC (500-600 USD/ton CO₂) is too expensive, mainly due to operational expenses. Although the distinct costs associated with active capture (e.g., the operation of fans, etc.) are presumably low, the costs are relatively fixed since fan technology has probably reached its technical limit. On the other hand, passive DAC devices can have lower operating expenses but tend to have a larger cross-sectional area than active devices, since they need to expose a larger surface area of sorbent to compensate for the intermittent nature of ambient air flows.

Direct air capture generally constitutes both CO₂ capture and its release. The release process is considered the primary economic limiting factor, driven by either a temperature, pressure, moisture, electrochemical gradient, or combination. In particular, the moisture-swing process requires relatively minor amounts of energy, in which the energy to release the CO₂ is the latent heat of the water. However, this process consumes significant amounts of water relative to the capture tonnage.

The limitations of active capture (i.e., fan driven capture) is the cost of the fans and motors and their operational cost. The limitation of passive capture is the cost of the required increase in surface area of exposed sorbent. The benefits of each method come at a cost.

Contemplated herein are systems and methods for passive carbon dioxide collection using an enhanced air flow. The system contemplated herein provides the benefits of conventional passive and active DAC technology without the high cost of active air movers or the low output of natural airflows. The contemplated CO₂ collection systems include a chimney or tower as an architectural addition that is able to harness the chimney or stack effect and provide an enhanced airflow, without requiring fans or other air moving devices. The chimney effect allows for more rapid CO₂ uptake without incurring the costs of an active system. With proper placement of the sorbent material, this chimney effect airflow will increase the airflow across the sorbent, as well as make it less intermittent, thus increasing the capture of CO₂ per unit time. While the chimney would add a modest upfront capital cost, it is able to function without additional

operating costs, thus removing the operational expense that hinders the development of active capture devices. According to various embodiments, the contemplated system can achieve the absorption speed of an active system while having the operating costs of a passive system.

The contemplated passive DAC system (hereinafter “chimney DAC system” or simply “system”) and method contemplated herein has the potential to disrupt the existing DAC technologies. Application of the chimney-like structure and careful attention to its details (e.g., shape, material, color, sorbent properties, sorbent placement, flow patterns, solar heating, etc.) enhances the limited airflow within an otherwise passive DAC setup, while avoiding the energy costs associated with active capture.

The contemplated systems and methods offer numerous advantages over conventional DAC systems, both passive and active. The chimney effect provides increased airflow compared to passive DAC technology, permitting the use of sorbent materials having faster collection rates than would be practical in passive systems. According to various embodiments, the system uses less energy per ton of captured CO₂ than devices that employ fans or other mechanical means to actively move air.

According to various embodiments, the contemplated system may provide additional energy savings and performance boosts by harnessing natural heating (e.g., solar, geothermal, etc.). In some embodiments, the materials used in the chimney and/or outer coatings or solar devices may facilitate solar heating. This heat may be used to enhance the operation of a sorbent regeneration system, and may also be used to enhance the airflow through the chimney, as will be discussed in greater detail, below.

The chimney also provides an advantageous platform on which solar panels may be mounted. Utilizing the vertical scale of the chimney, solar power can be used to operate the system without substantially increasing its footprint, permitting multiple systems to be positioned closer to each other than otherwise possible.

The contemplated system is also advantageous over conventional DAC systems, particularly active DAC systems, because it has few moving parts, reducing the need for maintenance and increasing operational safety without sacrificing performance. In some embodiments, the system may be automated, allowing for minimal staffing and further reducing operational costs. Remote monitoring and metering can provide certifiable proof of capture for either disposal or carbon re-use, according to various embodiments.

FIGS. 1A-1C are perspective, side, and cross-sectional views of a non-limiting example of a carbon dioxide collection system 100 with enhanced air flow. As shown, the system 100 comprises a chimney 106, a capture structure 104, and a release enclosure 102. According to various embodiments, the chimney 106 causes the creation of an airflow 108 which is used to expose the capture structure 104 to a large volume of atmosphere before being regenerated in the release enclosure 102. Each element will be discussed in greater detail below.

In the context of the present description and the claims that follow, a capture structure 104 is a structure that comprises a sorbent material 110, and is configured to expose that sorbent material 110 to an airflow 108 containing carbon dioxide 218. The capture structure 104 provides a structure or framework to position the sorbent material 110 used to capture, and subsequently release, atmospheric carbon dioxide 218, in an arrangement conducive for these two activities. The capture structure 104 is movable between a collection configuration 116 and a release configuration 208.

The capture structure 104 shown in FIGS. 1A-1C is in the collection configuration 116. These configurations will be discussed further in the context of FIGS. 2A and 2B, below.

According to various embodiments, the contemplated system 100 may be adapted for use in a wide range of environments, using a variety of sorbent materials 110 and capture structures 104 that meet the needs of a particular location. In some embodiments, including the non-limiting example shown in FIGS. 1A-1C, the capture structure 104 may comprise a plurality of vertical panels 112 (i.e., larger vertically than in either horizontal direction), each vertical panel 112 comprising a sorbent material 110. In some embodiments, these vertical panels 112 are oriented radially about a central axis 118 of the capture structure 104, such that they are pointing away from the central axis 118.

It should be noted that the plurality of radially-oriented vertical panels 112 described above and shown in FIGS. 1A-1C are but one of a wide range of capture structures 104 that may be used with the contemplated system 100. In some embodiments, the sorbent material 110 of the capture structure 104 may have rotational symmetry about the central axis 118, while in other embodiments the sorbent material 110 may be distributed across the capture structure 104 unevenly. In some embodiments the capture structure 104 may be a single structure, while in other embodiments the capture structure 104 may comprise a plurality of segments or components that are able to move independently. In some embodiments, the capture structure 104 may employ solid sorbent material 110 while in others it may use liquids.

In some embodiments the sorbent material 110 may be releasably coupled to the capture structure 104, able to be replaced or swapped with a different sorbent material 110. In other embodiments, the sorbent material 110 may be integral with the capture structure 104. In some embodiments, the capture structure 104 and its sorbent material 110 may be configured to collect carbon dioxide 218 as the airflow 108 created by the chimney 106 passes over an outer surface of the sorbent material 110. In other embodiments, the capture structure 104 and sorbent material 110 may be configured such that atmospheric carbon dioxide 218 is collected from the airflow 108 as it passes through the sorbent material 110. Additional variations of the capture structure 104 will be discussed below.

The system 100 also comprises a release enclosure 102. In the context of the present description and the claims that follow, a release enclosure 102 is a volume within which the sorbent material 110 of the capture structure 104 may be regenerated, causing the captured carbon dioxide 218 to be released into the enclosure 102, where it will be extracted for downstream storage, refinement, and/or use. According to various embodiments, the release enclosure 102 comprises a sorbent regeneration system, which is able to cause the sorbent material 110 of the capture structure 104 to release captured CO₂ for harvest, preparing the sorbent material 110 for another cycle of capture and release. In some embodiments, the release enclosure 102 may also comprise an opening 204 through which at least a portion of the capture structure 104 may be lowered to regenerate the sorbent material 110. In some embodiments, the release enclosure 102 may further comprise a lid 206 or other means by which the enclosure may be sealed. The release enclosure 102, the sorbent regeneration system, and the opening will all be discussed in greater detail with respect to FIGS. 2A and 2B, below.

The system 100 also comprises a tower or chimney 106. In the context of the present description and the claims that follow, a chimney 106 is a conduit sized and shaped such

that an airflow **108** is induced upward through said conduit by the stack effect or chimney effect. A pressure differential between the air inside the chimney **106** and outside the chimney **106** causes this airflow **108**. In some cases, this pressure differential may be generated by an air speed differential imposed by wind flow. In other cases, the airflow **108** may be caused by a temperature differential between the air inside the chimney **106** and outside the chimney **106**, or even a thermal gradient within the chimney **106**.

According to various embodiments, the chimney **106** is positioned such that the resulting airflow **108** passes through the capture structure **104** before exiting the chimney **106** while the capture structure **104** is in the collection configuration **116**. In some embodiments, the chimney **106** is positioned above the release enclosure **102**, such that the capture structure **104** may simply be raised up from and lowered down into the release enclosure **102** as it moves between the collection and release configurations. In other embodiments, the release enclosure **102** may be located elsewhere, or may even be intermittent, temporarily forming for regeneration and harvest, then breaking apart for collection.

A shape common to cooling towers is the hyperboloid, specifically a single sheet hyperboloid. The hyperboloid shape provides good structural strength using a minimal amount of building material. These shapes also advantageously accelerate the airflow **108** upward as it passes through the narrowed waist. According to various embodiments, the chimney **106** design can acknowledge and adjust for a number of variables including, but not limited to, system dimensions, location, and supporting structures. For example, the chimney **106** may be a hyperboloid structure **114** (i.e., having a hyperboloid shape), may be placed on the side of a hill, may use an underground/above ground combination for an additional temperature-driven effects, and the like.

According to various embodiments, the chimney **106** may have a circular cross-section, allowing capture to be “the same” regardless of the direction of the wind or other environmental air flow. Circular design for the chimney **106** makes the system **100** agnostic to directional consideration and uses the structural shape most consistent with strength for a raised element. However, other embodiments may employ a chimney **106** with a cross-section having a lower order of symmetry.

The ideal position for the chimney **106**, relative to the capture structure **104** when in the collection configuration **116**, depends upon the induced airflow **108**. According to various embodiments, the chimney **106** is positioned such that the airflow **108** passes through the capture structure **104** while in the collection configuration **116**. In some embodiments, the capture structure **104** in the collection configuration **116** may be below the chimney **106**, as shown in FIGS. 1A-1C. As an option, in some cases, the capture structure **104** may surround that area below the chimney **106**. In other embodiments, the capture structure **104** in the collection configuration **116** may be, at least partially, inside of the chimney **106**, as shown in FIG. 2A.

Ideally, the system **100** should be balanced such that the rate at which it is able to capture carbon dioxide **218** (i.e., CO₂ capture rate **124**) is equal to or substantially equal to the rate at which CO₂ is passing through the capture structure **104** inside the induced airflow **108** while the capture structure **104** is in the collection configuration **116** (i.e., CO₂ exposure rate **126**). The chimney **106** geometry and scale, and the capture structure **104** geometry and scale, are preferably chosen to expose the sorbent material **110** as

effectively as possible to the airflow **108** during capture and the regeneration media while in the release enclosure **102**. Moving air through the structure faster than the CO₂ can be captured is wasted energy, and moving it too slow is wasted potential.

Flow rate also should take into account the specific geometry of the capture structure **104**. For example, in one embodiment having vertical panels **112** of sorbent material **110** oriented radially away from a central axis **118**, the geometry of the capture structure **104** should be chosen such that the sorbent material **110** at the backend of the airflow **108** gets exposure for maximized capture during the cycle time.

The CO₂ capture rate **124** is mainly influenced by the sorbent material **110** being used, as well as ambient environmental conditions **128**, as they pertain to the “swing” of that particular sorbent material **110** (e.g., humidity, rain, temperature, etc.). The CO₂ exposure rate **126** is mainly influenced by the geometry of the chimney **106** as well as the geometry and position of the capture structure **104**, relative to the chimney **106**. According to various embodiments, these two rates may be tuned to be substantially equal through modification of one or both of the structures and/or the sorbent material **110** used.

According to various embodiments, the geometry of the capture structure **104** may be chosen such that, for a given chimney **106** and resulting airflow **108** as well as a sorbent material **110** chosen for the intended environment using the available resources, the CO₂ capture rate **124** (i.e., CO₂ captured by capture structure **104** per unit time) due to that sorbent material **110** and how it is arranged in the capture structure **104** is substantially equal to the CO₂ exposure rate **126** (i.e. CO₂ passing through and contacting the capture structure **104** per unit time) caused by the airflow **108** and the overall size of the capture structure **104**. Overall, modifying the distance of the capture structure **104** in the collection configuration **116** from the chimney **106** can help balance these rates. The further away from the chimney **106**, the lower the pressure drop, and the lower the speed of airflow **108**.

As a specific, non-limiting example, in some embodiments the collection configuration **116** comprises at least one of the capture structure **104** being separated from the chimney **106** by at least a vertical separation **120** (e.g., the vertical separation between the top of the capture structure **104** and the bottom of the chimney **106**, etc.) and the sorbent material **110** of the capture structure **104** being distanced from a central axis **118** of the capture structure **104** by at least a radius **122**. See, for example, the side view of the system **100** shown in FIG. 1B and the cross-section taken along A-A of FIG. 1B, shown in FIG. 1C. According to various embodiments, at least one of the vertical separation **120** and the radius **122** may be chosen such that the CO₂ capture rate **124** of the sorbent material **110** as arranged in the capture structure **104** is substantially equal to the CO₂ exposure rate **126** caused by the airflow **108** passing through the capture structure **104** while the capture structure **104** is in the collection configuration **116**. It may be easier to fine tune the balance between exposure and capture through modifying one or more of these distances, as changes in the choice of sorbent material **110** may modify that balance too drastically for a particular installation.

In some embodiments, manipulating the distance between the capture structure **104** (or more specifically, the sorbent material **110** of the capture structure **104**) and the chimney **106** may be used to balance the exposure and capture rates at the time of installation of the system **100**. In other

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embodiments, the capture structure **104** may be configured such that at least one of said distances may be changed after installation of the system **100**. For example, in one embodiment, the relative position of the capture structure **104** with respect to the chimney **106** may be changeable through a mechanism (e.g., hydraulics, linear actuator, winch, etc.), allowing the system **100** to adapt to a change in an environmental condition **128**. Examples of a change in an environmental condition **128** include, but are not limited to, changes in weather (e.g., temperature, humidity, cloud cover, rain, wind, etc.), changes in consumables (e.g., introduction of a contaminant to a regeneration media, etc.), and the like. In some embodiments, the system **100** may be configured to detect a change in an environmental condition **128** and automatically modify the geometry of the capture structure **104** in response, to maintain efficiency.

According to various embodiments, the system **100** may comprise other sub-systems not shown in FIGS. 1A-1C including, but not limited to, one or more product outlets and outflow piping for obtaining a product stream (e.g., CO₂-enriched air or other gas, CO₂-enriched liquid, etc.), a control system, sensors, mechanisms for moving the capture structure **104** between the collection and release configurations (e.g., lifts, cranes, actuators, etc.). In some embodiments where the regeneration of the sorbent material **110** requires the use of a limited regeneration media such as clean water, the system **100** may be coupled to a source of replacement media. In some embodiments the system **100** may be self-powered by on-board solar cells, wind turbines, hydroelectric generators, and the like. In other embodiments, the system **100** may be coupled to external power sources. In some embodiments, the system **100** may also be coupled to infrastructure for the downstream handling of the product stream (e.g., purification, storage, use as feedstock, compression, etc.).

FIGS. 2A and 2B are cross-sectional views taken along the central axis **118** of a non-limiting example of an embodiment of the contemplated carbon dioxide collection system **200**, with a capture structure **104** in the collection and release configurations, respectively. As shown, the release enclosure **102** comprises a sorbent regeneration system **202** and an opening **204** sized to receive at least a majority of the capture structure **104** (e.g., in an embodiment where the regeneration media **210** is liquid water, the entire capture structure **104** would not have to be submerged, just the sorbent material **110**).

According to various embodiments, the contemplated systems operate by cycling between collection and release phases. According to various embodiments, the capture structure **104** has a release configuration **208** and a collection configuration **116**. In the context of the present description and the claims that follow, a collection configuration **116** is an arrangement of the capture structure **104** such that the sorbent material **110** is exposed to an airflow **108** (i.e., the airflow **108** generated by the chimney **106**), allowing the sorbent material **110** to capture carbon dioxide **218** from the airflow **108**. As shown in FIG. 2A, in which the capture structure **104** is in the collection configuration **116**, the capture structure **104** is elevated above the release enclosure **102**.

In some embodiments, the release enclosure **102** may be positioned directly beneath the capture structure **104** such that at least the portion of the capture structure **104** comprising the sorbent material **110** may be lowered into the enclosure for harvesting. In other embodiments, the release enclosure **102** may be located elsewhere. A particular

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embodiment, where the release enclosure **102** is intermittent, will be discussed with respect to FIG. 11, below.

In some embodiments, the release enclosure **102** may be a single container or basin, able to receive all of the sorbent material **110** of the capture structure **104**. See, for example, the release enclosure **102** shown in FIGS. 1A-1B. In other embodiments, the release enclosure **102** may be partitioned into segments that may more closely fit a segmented geometry of some embodiments of the capture structure **104**, thereby reducing the overall volume contained within the release enclosure **102** and reducing wasted regeneration media **210** (e.g., thermal energy, water, energy needed to reduce pressure, etc.). As a specific, non-limiting example, the capture structure **104** shown in FIGS. 1A-1C comprises six vertical panels **112**, each having sorbent material **110**. In one embodiment (not shown), each of those six vertical panels **112** may have a single-panel-sized vessel into which it is lowered, minimizing the total volume of the release enclosure **102** while still able to regenerate the entire capture structure **104** at the same time.

In some embodiments where the release enclosure **102** comprises a plurality of vessels, the vessels may be in fluid communication with each other, facilitating the dispensing of regeneration media **210** and the extraction of a product stream **216** of CO₂-enriched fluid **212**. In other embodiments, each vessel may be in fluid communication with the sorbent regeneration system **202** and a product outlet **214** without being directly connected to other vessels. In some embodiments, a compromise between these two extremes may be a preferred approach.

The release enclosure **102** shown in FIGS. 2A and 2B is also a single vessel, in this case ring shaped. The ring shape results in a release enclosure **102** volume that is smaller than that of FIG. 1A, but using a single vessel provides advantages when the enriched fluid **212** is a gas that is being removed using a sweep gas, for example.

In some embodiments, the sorbent material **110** remains outside of the chimney **106** in both the collection configuration **116** and the release configuration **208**. In other embodiments, including the non-limiting example shown in FIG. 2A, the collection configuration **116** may comprise the capture structure **104** being at least partially inside of the chimney **106**. As previously discussed, the efficiency of the system may be tuned through changing the distance between the capture structure **104** and the chimney **106**. As an added bonus, placing at least part of the capture structure **104** inside the chimney **106** while in the collection configuration **116** may provide protection for the sorbent material **110**, which is sometimes fragile.

Once an adequate amount of CO₂ is collected, the capture structure **104** moves into the release configuration **208**, where at least a portion of the capture structure **104** comprising the now CO₂-laden sorbent material **110** is placed inside the release enclosure **102**. In the context of the present description and the claims that follow, a release configuration **208** comprises the capture structure **104** being sufficiently enclosed inside the release enclosure **102** that the regeneration media **210** of the sorbent regeneration system **202** may operate on the sorbent material **110** of the capture structure **104** to release carbon dioxide **218** collected by the capture structure **104** while in the collection configuration **116** to form an enriched fluid **212** within the release enclosure **102**.

The regeneration media **210** will vary depending on the nature of the sorbent material **110** used. Examples include, but are not limited to, liquid, vapor, gas, heat, vacuum or reduced pressure, electricity, water, microwaves, and the

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like. Exposure to the regeneration media **210** will cause the captured carbon dioxide **218** to be released into the release enclosure **102**. In some embodiments, the carbon dioxide **218** is released into the regeneration media **210**, where it may be stored. If the regeneration media **210** can retain the carbon dioxide **218** released by the sorbent material **110**, in ambient conditions, the release enclosure **102** may remain open and may not require a lid **206** or be otherwise sealed.

In some embodiments, the release enclosure **102** may comprise an opening **204** through which at least a portion of the capture structure **104** may be lowered to regenerate the sorbent material **110**. In some embodiments, the release enclosure **102** may further comprise a lid **206** sized to cover the opening **204** and configured to seal the capture structure **104** inside the release enclosure **102** when in the release configuration **208**. Such a sealing would prevent or significantly reduce the amount of released carbon dioxide **218** and/or regeneration media **210** (e.g., steam, low pressure environment, heat, moisture, etc.) from escaping. In other embodiments, the release enclosure **102** may remain open, and rely on a sorbent regeneration system **202** that is not disrupted by exposure to the atmosphere (e.g., submerging the sorbent in a liquid and releasing the captured CO₂ as dissolved inorganic carbon, etc.).

The carbon dioxide **218** is then removed from the release enclosure **102** through a product outlet **214** as a product stream **216** of enriched fluid **212** (e.g., CO₂-enriched air or other gas, CO₂-enriched liquid, etc.). The product outlet **214** is in fluid communication with the inside of the release enclosure **102**. This product stream **216** may then move on to downstream processing and/or use. After the sorbent regeneration is complete and there is no danger of losing meaningful amounts of collected CO₂, the capture structure **104** moves back into the collection configuration **116** and the process begins again. In other embodiments, the system may operate in a continuous fashion, collecting carbon dioxide **218** and producing a product stream **216** simultaneously. This will be discussed in greater detail with respect to FIG. 9, below.

The chimney effect may be achieved with a variety of conduit geometries. This advantageously allows the chimney **106** of the contemplated system **100** to be adapted to take advantage of aspects of the location where it will be deployed and the environment in which it will operate (e.g., temperature, airflow, sun exposure, etc.). For example, a chimney **106** may be modified to operate on a hillside.

In some embodiments, the chimney **106** may be manufactured specifically for the contemplated system **100**. In other embodiments, the system **100** may repurpose conduits built for other purposes. For example, in one embodiment, old cooling towers and smokestacks may be retrofit for use as part of a carbon dioxide collection system **100** with enhanced airflow. In another embodiment, the cooling tower of a HVAC system may be leveraged for creating an enhanced airflow **108**. In still other embodiments, the system **100** contemplated herein may be integrated with cooling towers, smokestacks, and the like that are still in active use. As an option, the system **100** may be integrated to harness aspects of these active industrial systems (e.g., using waste heat to enhance a temperature differential and improve airflow **108**, utilize infrastructure already present such as water and electricity, etc.). Those skilled in the art will recognize that the chimney effect has been utilized in a wide range of industries and technologies, presenting numerous opportunities for repurposing chimneys and towers for the task of capturing atmospheric carbon dioxide **218**.

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One of the advantages of the contemplated system **100** is that it may be implemented across a wide range of scales. In some embodiments, the chimney **106** may be sized such that it may be mass-produced in a factory, and transported to the installation site on a tractor trailer without requiring special permits. Other embodiments may be the size of a shipping container. The ability to mass produce the chimney **106** inside a factory minimizes the cost, improves the quality, and facilitates rapid worldwide distribution. The ability to scale the system **100** small also opens a wide range of locations and applications.

Of course, in other embodiments, the chimney **106** may be too large to manufacture off-site. However, geometries may be employed that reduce the cost and complexity of an on-site construction. A shape common to cooling towers is the hyperboloid, specifically a single sheet hyperboloid.

FIG. 3 shows a perspective view of a non-limiting example of a chimney **306** having a hyperboloid structure **114** and dynamic geometry **300**. The hyperboloid structure **114** provides good structural strength using a minimal amount of building material. These shapes also advantageously accelerate the airflow **108** upward as it passes through the narrowed waist. Furthermore, the hyperboloid structure **114** may be constructed using straight members **308**, as shown in FIG. 3, rather than requiring curved beams that would be more difficult and expensive to manufacture and transport to the installation site. The hyperboloid shape has benefits that apply to large implementations that are also available when implemented on a much smaller scale.

Like most other implementations of chimneys and towers, in some embodiments, the contemplated chimney **106** geometry may remain fixed. However, in other embodiments, the chimney **306** size and shape may be dynamic. In the context of the present description and the claims that follow, a chimney **306** having a dynamic geometry **300** comprises at least one structural attribute (e.g., height **302**, width **304**, etc.) that can be modified after the chimney **306** has been manufactured.

As a specific example, the chimney **306** shown in FIG. 3 shows two arrows indicating directions of relative rotation. Rotating the openings of this chimney **306** relative to each other will cause the mouths to expand or contract (e.g., width **304**) and the chimney **306** to grow taller or shrink (e.g., height **302**), assuming the rigid straight members **308** are coupled to each other with a material that is deformable or elastic, or with a structure or framework that is configured to expand and contract as needed. Other embodiments may comprise chimneys **306** with dynamic geometry **300** that operate on other principals or having other shapes.

Such a dynamic chimney **306** may be advantageous for use in environments with wildly changing conditions, because changing the height **302**/width **304** of a hyperboloid chimney **306** directly impacts the speed and volume of air moving through it. A change in shape may be a very energy efficient way to adjust the speed of the airflow **108** in response to a change in an environmental condition **128** (e.g., temperature, humidity, exposure to sunlight, etc.). In other embodiments, a dynamic geometry **300** such as these may be mass produced, which would provide better quality at a lower cost, and still allow a chimney **306** to be fine-tuned at installation to induce an airflow **108** ideal for a particular capture structure **104** having a particular sorbent material **110** operating at a particular location.

FIG. 4 is a perspective view of a non-limiting example of a carbon dioxide collection system **400** using solar heating. As previously mentioned, the advantages the contemplated system **100** offers over conventional DAC systems stem

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from the airflow **108** induced by the chimney **106**. The airflow **108** is caused, at least in part, by the temperature differential between the air inside the chimney **106** and the air outside. Thus, that airflow **108** may be improved by increasing the difference in temperature. According to various embodiments, the construction material of the chimney **106** may be highly heat-conductive to enhance airflow **108** further. According to various embodiments, a chimney **106** may comprise a heating material **402** configured to absorb solar radiation **404** and heat the chimney **106** such that the air temperature **406** inside the chimney **106** is greater than the ambient air temperature **408** outside the chimney **106**.

In some embodiments, the outside of the chimney **106** may be colored or otherwise coated with a heating material **402** to enhance the airflow **108** through the temperature differential. In the context of the present description and the claims that follow, a heating material **402** is a material having properties to assist with heating the chimney **106**, whether through absorbing solar radiation **404** or through other means.

According to various embodiments, application of solar photovoltaic (PV) panels **410** to the chimney's exterior may also be helpful to provide electrical energy to power various aspects of the system **400**. As an option, the system **400** may still be connected to a local power grid for back-up power. In some embodiments, since the airflow **108** does not require any power, the majority, if not all, of electricity consumed by the system **400** may be for regeneration and compression. Implementation of solar panels **410** could reduce the energy costs and balance the energy consumption, though solar may not be appropriate for all environments. Attachment of solar panels **410** to the chimney's exterior could increase the solar panel **410** efficiency and life span of the photovoltaic cells as the airflow **108** cools the panels. Furthermore, solar panels **410** may add heat at strategic locations to enhance the airflow **108** like the previously discussed heating material **402**.

FIG. **5** is a cross-sectional view along the central axis **118** of a non-limiting example of a chimney **500** using a heating material **402** to create a thermal gradient **506**. According to various embodiments, heating material **402** (e.g., black paint, etc.) may be used to heat some portions of the chimney **500** and/or cooling material (e.g., white paint, etc.) may be used to cool others, creating a thermal gradient **506**. As an option, the color coating or other coating materials may be placed in a pattern or a mix of colors and patterns to achieve the desired heat and airflow **108**. In some embodiments, an upper portion **502** of the chimney **106** may be coated with a heating material **402** so the upper portion **502** absorbs more solar radiation **404** than the lower portion **504**, creating a thermal gradient **506** that acts like the fan in an induced mechanical draft cooling tower, which produces an exit air velocity that is higher than the entrance air velocity, enhancing the overall airflow **108** through the capture structure **104**.

In some embodiments, the sorbent regeneration media **210** (e.g., water, vapor, liquid, gas, etc.) may use the collected heat to speed up the regeneration process, leading to a quicker harvest and shortened collect/release cycle. As a specific example, in one embodiment, a thin tank full of regeneration media **210** or other fluid may line the chimney **106** to capture heat for use in the sorbent regeneration system **202**. In another embodiment, the lower portion **504** of the outside of the chimney **106** may have tubes that heat and deliver water for use by the regeneration process while also shuttling away heat from the lower portion **504** of the chimney **106**, enhancing the thermal gradient **506**.

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Some embodiments of the system **500** may use the most energy-promising sorbent material **110**/release mechanism, moisture-swing. However, with the addition of heat gathered from the chimney **500**, thermal-swing materials might be equally attractive and avoid the typical associated external heating costs. In some embodiments, the heat may also be used to increase the storage efficiency of the captured CO₂.

In one embodiment, the temperature of the chimney **500** may be manipulated to improve the airflow **108** by positioning a portion of the system **100** below grade to create additional temperature differential. In another embodiment, a hillside might be used to take advantage of natural flows of air to passively achieve a greater flow of air through the sorbent, thus "speeding-up" capture. The achievement of an increased airflow **108** to allow the sorbent to have the opportunity to capture more CO₂, coupled with a sorbent material **110** that has a capture absorption rate that is matched to the airflow **108**, can lead to greater output. Faster capture and harvest reduces the equipment cost per ton of CO₂.

FIG. **6** shows a non-limiting example of a portion of a capture structure **600**. Specifically, it shows half of a rack structure to which sorbent materials **110** would be attached. According to various embodiments, this rack could be lifted up for exposure to the airflow **108**, and then lowered into the release enclosure **102** for harvest. According to various embodiments, this rack-based geometry could be used inside and outside the chimney **106**. In some embodiments, it may be roughly the size of the bottom of the chimney **106**, having a radius **122** and separated from the chimney **106** by a height **302**, as shown in FIGS. **1B** and **1C**. In some embodiments, the capture structure **104**, release enclosure **102**, and the chimney **106** may all share the same central axis **118**. Other embodiments may have less symmetry, due to the specific environment where it is to be installed.

According to various embodiments, the capture structure **104** comprises some sort of structure configured to support and arrange a sorbent material **110** such that it has efficient exposure to the airflow **108** during collection and the regeneration media **210** during release. There are numerous ways a sorbent material **110** may be attached to a structure such as the rack shown in FIG. **6**. FIGS. **7-9** show non-limiting examples of three different arrangements, which will be discussed below. It should be noted, however, that there are many other ways the sorbent material **110** may be supported. Those skilled in the art will recognize that the capture structures **104** being discussed here may be adapted for use with a wide range of materials, and that the contemplated system **100** may comprise capture structures **104** adapted from other DAC devices known in the art, both passive and active.

FIG. **7** is a top view of a non-limiting example of a capture structure **700** having a plurality of sorbent panels **702** attached to a capture rack **704**. As shown, the sorbent panels **702** are oriented radially about a central axis **118** (i.e., the central axis **118** of the capture structure **700**). This orientation is ideal as it is able to receive airflow **108** coming from any and all directions as it moves inward and upward through the chimney **106** above. These sorbent panels **702** comprise sorbent material **110**. As shown, the capture structure **104** is constructed as an open cylinder, allowing the airflow **108** to move inward and upward freely, after passing through and contacting the sorbent panels **702**.

FIG. **8** is a perspective view of another non-limiting example of a capture structure **800**, this one having a plurality of horizontal rings **802**. The rings are fastened to a collection of columns (not shown) that function like the rack

of FIG. 7. In some embodiments, these rings **802** may be porous, allowing air to flow through them and the sorbent material **110** they contain, in addition to contacting the sorbent material **110** on their outer surface. FIG. 9 is a side view of another non-limiting example of a capture structure **900**, this having a plurality of horizontal disks **902** with a central hole **904** to allow the airflow **108** to move inward and upward.

As previously stated, in some embodiments, it may be desired to have the sorbent material **110** even closer to the chimney **106** where the airflow **108** may be moving at a speed ideal for that particular set of conditions (e.g., sorbent material **110**, environmental conditions **128**, air speed, etc.). FIG. 10 shows a non-limiting example of a system **1000** with a capture structure **1002** configured for use inside of a chimney **106**. This capture structure **1002** is also able to collect and harvest simultaneously, providing a continuous product stream **216**. This is possible because the sorbent material **110** being used is a liquid sorbent **1004** that circulates up and down the capture structure **1002**, capturing CO_2 from the air as it flows upward through the chimney **106**. The CO_2 -laden liquid sorbent **1004** circulates back down to the release enclosure **102** at the bottom, where the sorbent is regenerated and the carbon dioxide **218** extracted, as is known in the art. In some embodiments, a chimney **106** may have a single capture structure **104** inside of it, while in other embodiments the chimney **106** may have multiple capture structures **104**.

FIG. 11 is a non-limiting example of another capture structure **1100** configured for use within a chimney **106**. The shaded panels rotate, with one side exposed to the airflow **108** and the other facing inward. The space inside the capture structure **1100** becomes an intermittent release enclosure **1102** after the panels rotate. Laden sorbent lines the walls of the intermittent release enclosure **1102** facing inward. The regeneration media **210** (i.e., heat) causes the CO_2 to be released inside the enclosure **1102**. As an option, the airflow **108** itself may be used as a sweep gas to move the released CO_2 out of the enclosure **1102** and into downstream processing (not shown). The panels then rotate and the cycle continues. This embodiment is somewhere between the batch operation of the panel embodiments and the continuous operation discussed with respect to FIG. 10.

Advantageously, the systems and methods contemplated herein may be adapted for use with any sorbent material **110** able to be regenerated. This flexibility extends the range of operating conditions in which the system **100** may be deployed, in terms of weather and climate. Examples include, but are not limited to, swings based on temperature, pressure, moisture, electrochemical gradient, electricity, and the like. As previously discussed, in some embodiments, heat may be drawn from the chimney **106** to enhance the regeneration process. In some embodiments, the sorbent material **110** may be solid, while in others it may be liquid. Those skilled in the art will recognize that the contemplated system **100** and method may be adapted for use with any form factor these sorbent materials **110** may be given.

It will be understood that implementations are not limited to the specific components disclosed herein, as virtually any components consistent with the intended operation of a method and/or system for passive collection of atmospheric CO_2 may be utilized. Accordingly, for example, although particular chimneys, capture structures, sorbent materials, release enclosures, and configurations may be disclosed, such components may comprise any shape, size, style, type, model, version, class, grade, measurement, concentration, material, weight, quantity, and/or the like consistent with the

intended operation of a method and/or system for the passive collection of atmospheric CO_2 may be used. In places where the description above refers to particular implementations of a system for passive collection of atmospheric CO_2 , it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these implementations may be applied to other passive collection systems.

What is claimed is:

1. A carbon dioxide collection system, comprising:

a release enclosure comprising an opening and a sorbent regeneration system;

a capture structure comprising a plurality of vertical panels oriented radially about a central axis of the capture structure, each vertical panel comprising a sorbent material, the capture structure being movable between a collection configuration and a release configuration;

a chimney having a hyperboloid structure and shaped such that an airflow upward through the chimney is created, the chimney positioned above the release enclosure such that the airflow passes through the capture structure before exiting the chimney while the capture structure is in the collection configuration, the chimney comprising a heating material configured to absorb solar radiation such that an air temperature inside the chimney is greater than an ambient air temperature outside the chimney;

at least one solar panel coupled to the chimney; and

a product outlet in fluid communication with the inside of the release enclosure and configured to emit a product stream of enriched fluid;

wherein the collection configuration comprises the capture structure elevated above the release enclosure such that the sorbent material of the capture structure is exposed to the airflow generated by the chimney, allowing the sorbent material to capture carbon dioxide from the airflow; and

wherein the release configuration comprises the capture structure being sufficiently enclosed inside the release enclosure that the sorbent regeneration system may operate on the sorbent material of the capture structure to release carbon dioxide collected by the capture structure while in the collection configuration to form an enriched fluid within the release enclosure.

2. The carbon dioxide collection system of claim 1:

wherein the collection configuration comprises at least one of the capture structure being separated from the chimney by at least a vertical separation and the sorbent material of the capture structure being distanced from a central axis of the capture structure by at least a radius; and

wherein at least one of the vertical separation and the radius is chosen such that a CO_2 capture rate of the sorbent material as arranged in the capture structure is substantially equal to a CO_2 exposure rate caused by the airflow passing through the capture structure while the capture structure is in the collection configuration.

3. The carbon dioxide collection system of claim 2, wherein the at least one of the radius and the vertical separation can be modified after the carbon dioxide collection system has been installed, in response to a change in an environmental condition.

4. The carbon dioxide collection system of claim 1, wherein the release enclosure further comprises a lid sized

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to cover the opening and configured to seal the capture structure inside the release enclosure when in the release configuration.

5. The carbon dioxide collection system of claim 1, wherein the chimney is one of a cooling tower and a smokestack.

6. The carbon dioxide collection system of claim 1, wherein the chimney comprises a dynamic geometry such that at least one of a height of the chimney and a width of the chimney may be modified after manufacture.

7. The carbon dioxide collection system of claim 1, wherein the heating material is confined to an upper portion of the chimney such that the upper portion absorbs more solar radiation than a lower portion of the chimney, thereby creating a thermal gradient to enhance the airflow through the chimney.

8. The carbon dioxide collection system of claim 1, wherein the collection configuration comprises the capture structure being at least partially inside of the chimney.

9. A carbon dioxide collection system, comprising:

a release enclosure comprising an opening and a sorbent regeneration system;

a capture structure comprising a sorbent material, the capture structure being movable between a collection configuration and a release configuration;

a chimney shaped such that an airflow upward through the chimney is created, the chimney positioned above the release enclosure such that the airflow passes through the capture structure before exiting the chimney while the capture structure is in the collection configuration; and

a product outlet in fluid communication with the inside of the release enclosure and configured to emit a product stream of enriched fluid;

wherein the collection configuration comprises the capture structure elevated above the release enclosure such that the sorbent material of the capture structure is exposed to the airflow generated by the chimney, allowing the sorbent material to capture carbon dioxide from the airflow; and

wherein the release configuration comprises the capture structure being sufficiently enclosed inside the release enclosure that the sorbent regeneration system may operate on the sorbent material of the capture structure to release carbon dioxide collected by the capture structure while in the collection configuration to form an enriched fluid within the release enclosure.

10. The carbon dioxide collection system of claim 9, wherein the capture structure comprises a plurality of vertical panels oriented radially about a central axis of the capture structure, each vertical panel comprising the sorbent material.

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11. The carbon dioxide collection system of claim 9:

wherein the collection configuration comprises at least one of the capture structure being separated from the chimney by at least a vertical separation and the sorbent material of the capture structure being distanced from a central axis of the capture structure by at least a radius; and

wherein at least one of the vertical separation and the radius is chosen such that a CO₂ capture rate of the sorbent material as arranged in the capture structure is substantially equal to a CO₂ exposure rate caused by the airflow passing through the capture structure while the capture structure is in the collection configuration.

12. The carbon dioxide collection system of claim 11, wherein the at least one of the radius and the vertical separation can be modified after the carbon dioxide collection system has been installed, in response to a change in an environmental condition.

13. The carbon dioxide collection system of claim 9, wherein the release enclosure further comprises a lid sized to cover the opening and configured to seal the capture structure inside the release enclosure when in the release configuration.

14. The carbon dioxide collection system of claim 9, further comprising at least one solar panel coupled to the chimney.

15. The carbon dioxide collection system of claim 9, wherein the chimney is one of a cooling tower and a smokestack.

16. The carbon dioxide collection system of claim 9, wherein the chimney is a hyperboloid structure.

17. The carbon dioxide collection system of claim 9, wherein the chimney comprises a dynamic geometry such that at least one of a height of the chimney and a width of the chimney may be modified after manufacture.

18. The carbon dioxide collection system of claim 9, wherein the chimney comprises a heating material configured to absorb solar radiation such that an air temperature inside the chimney is greater than an ambient air temperature outside the chimney.

19. The carbon dioxide collection system of claim 18, wherein the heating material is confined to an upper portion of the chimney such that the upper portion absorbs more solar radiation than a lower portion of the chimney, thereby creating a thermal gradient to enhance the airflow through the chimney.

20. The carbon dioxide collection system of claim 9, wherein the collection configuration comprises the capture structure being at least partially inside of the chimney.

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